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Assessment Task 1: Data Visualization Foundations

Report on Universal Store Holdings Limited (UNI)

Introduction

This report provides the complete analysis of the Universal Store Holdings Limited stock market during the fiscal year 2024-2025. This study uses different financial indicators like share price, trading volume, market capitalization, dividend yield, and price to earnings ratios to evaluate the company's fundamentals and market sentiment. Similarly, this study evaluates different portfolios under different strategies, such as buy and hold, single trade, and multiple trade scenarios. Likewise different type of visualization was used so that it can gain the attention to important trends, irregularities and breakthrough. This was done with the help of various charts such as dual axis chart, candlestick charts and annotated graphs. Overall, the main aim of the report is to give the investors a actional advice in terms of how strong visualization storytelling can help in decision making than comparing it to raw or AI generated outcomes.

Data Preparation and Collection Process

The Company's Universal Store Holdings Limited (UNI) stock market performance during the fiscal year 2024-2025 served as the basis for the dataset. A time series of datasets was created by compiling daily records of share prices, trading volume, and other financial variables. Data preparation included removing duplicates or missing values, standardizing units (for example, currency in dollars, yield in percentages, shares in absolute numbers), and aligning dates so that different variables could be accurately compared over the same time. To ensure consistency and transparency, portfolio value, dividend yield, price-to-earnings (P/E) ratio, and market were calculated.

Dataset Characteristics and Data Types

This UNI company dataset consists of various key financial and market metrics:

• Closing Price (numeric, currency $): It is a type of metric that is captured every day to monitor the UNI's share.

• Trading Volume (numeric, whole numbers): Trading volume is a type of metric that helps in knowing the shares traded each day and also reflects the activity of investors.

It is a type of metric that shows the shares traded each day, reflecting investor activity.

•Issued Shares (numeric, whole numbers): In the dataset, it is fixed at 76,720,810, and it is used to calculate market capitalization.

• Market Capitalization (numeric, currency $): It helps indicate the company value, which is calculated by price × shares on issue

• Dividend Yield (percentage %): Computed as dividend ÷ share price, expressing investor return on income.

• Price-to-Earnings Ratio (numeric, ratio): It is Fixed at 21.86, which is utilized for comparing valuation and market price.

•Portfolio Value (numerical representation, currency $): It is a type of metic which can be obtained from three different strategies like buy-and-hold, individual trade, and multiple trades – indicating fluctuations in value over a specified timeframe.

Overall, this data encompasses time series like daily prices and volumes, ratios and percentage yield, P/E, and absolute numbers like shares and portfolio balances.All this present the picture of the company's financial performance, investors' return and market behavior

1. Close Vs Volume

Findings from the Data and Charts

The relationship between trading activity volume and price movement is evident from the chart. A Significant finding near early September 2024 reveals that trading volume surged up to around 17.6 million shares right before the price peak. That is indicative of a surge in investor activity that has driven up demand and price. After that interval, the price settled down at that elevated base before touching another peak during early 2025. Significant trade volume dips are indicative of diminished market participation and a weaker selling market. The overall data reveals that a surge in the volume has generally preceded price peaks, and decreasing volume may be indicative of fading momentum in a declining market.

Visualization Types and Techniques

A dual-axis line chart was used, comparing the closing price (yellow line, right axis) with trading volume (blue line, left axis). By taking this format, market activity and price movement could both be equally compared without limiting the range of either of the data sets. Labelling techniques complemented storytelling:

• A blue arrow highlighted the high-volume spike followed by a price peak.

• A second arrow marked the decreasing volume with price drops, illustrating the link among trading activity, price movements, and trends.

• The highest volume quantity (1,763,867) was personally hand-written to draw attention to the anomaly.

The application of complementary colors (price in yellow, volume in blue) guaranteed that the distinction between the variables was clear, and the trendline smoothed out the series of volumes so that long-term trends became visible.

Challenges Encountered and Solutions

In this stock chart analysis, the main difficult part was axis scaling as the values of volume were much larger than prices, which would have buried the dataset if it was not placed along one axis. This problem was addressed by applying dual axis, while keeping both series readable. The second challenge was that of data overlapping, spikes in the volume obscuring other values; this was addressed by annotating only the most noteworthy points with annotations rather than overcrowding the chart. Lastly, data crossing raised the risk of confusion of the volume and the price, which was addressed by different colors and proper placement with their respective axes. These tweaks allowed the chart to efficiently communicate the relationship of the volume with the price, providing evidence in support of the observation that spikes of the volume are leading indicators of peak prices

2. Stock Chart

Findings from the Data and Charts

The UNI stock initially shows a strong upward trend which peaked at a breakout area in the $7.5-$8.0 range where price were pushed by momentum. Following the breakthrough period, there was a period of lateral trading which shows price consolidated until hitting a high of about $9.5 after which it marked a steep decline. After a strong rise the pullback suggests profit taking or

selling pressure after a steeper move up. Similarly, one of the noteworthy outliers is steeper upsurge at the breakout which was different from the earlier gradual up movement. Likewise, there are short-term corrections and fluctuations even when the trendline shows the long-term upward trajectory.

Visualization Types and Techniques

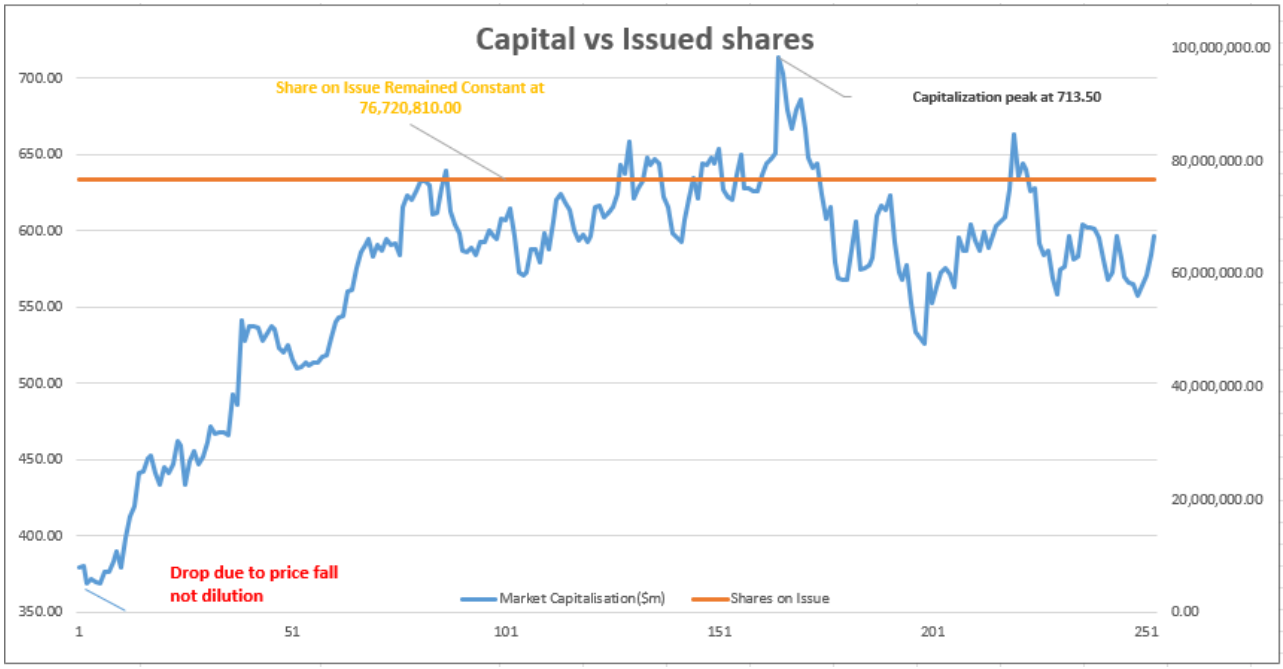
A candlestick chart was utilized, which is manly appropriate for illustrating intraday price change with Open,High,Low and Close (OHLC) values. This kind of chart makes it easy to identify key trends like pullbacks and breakouts. Likewise ,to make the chart more vivid and support storytelling.

- The red arrow indicated the subsequent reversal and pullback, while the green arrow indicated the breakthrough area, when prices were driven through resistance by positive emotion.
- The terms "Pullback" and "Breakout Zone" provided context for these important areas of price activity.
- Despite fluctuations, a straight trendline that represented the overall inclining tendency was included.

Problems Faced and Solutions

In the UNI dataset, data crowding and overlapping were the primary issues, as closely spaced candles can overcrowd the chart. By making the candle stick thin and highlighting only the visible locations with labels and arrows, this was avoided. Axis scaling was another problem which caused large price fluctuations, which pushed out smaller ones. To maintain the relative effects of price change, an even vertical scale was selected. Likewise, annotation was positioned carefully to make sure that there were no candlesticks covered. With these parameters, the chart was able to convey a clear story: breakout, then volatility, pullback, and continuing price action along an up-trending path.

3. Capital Vs Issued Shares



Results from the Data and Charts

The analysis of the UNI Issued Shares vs Capital chart provides that outstanding shares remained constant at 76,720,810, confirming no dilution nor new issues had taken place during the period. Although market capitalization was highly volatile, with the sharp earlier fall indicating a share price drop rather than share movement, the chart demonstrates peak capitalization at $713.50 million, showing healthy price growth at that point. This suggests that market capitalization changes were due entirely to price movements, rather than fundamental shifts in equity. The exception is the sharp earlier fall, which supports that the company valuation should fall rapidly without share structure movement.

Visualization Types and Techniques

It was decided that a dual-axis line chart should be used, with market capitalization (blue line) on one axis and shares on issue (orange line) on another. This was done so that the relationship

between the two measures - one variable and one constant - was readily shown. Labelling was used sparingly so that the results of most significant interest were highlighted:

• The number of shares in issue was emphasized by means of yellow highlighting.

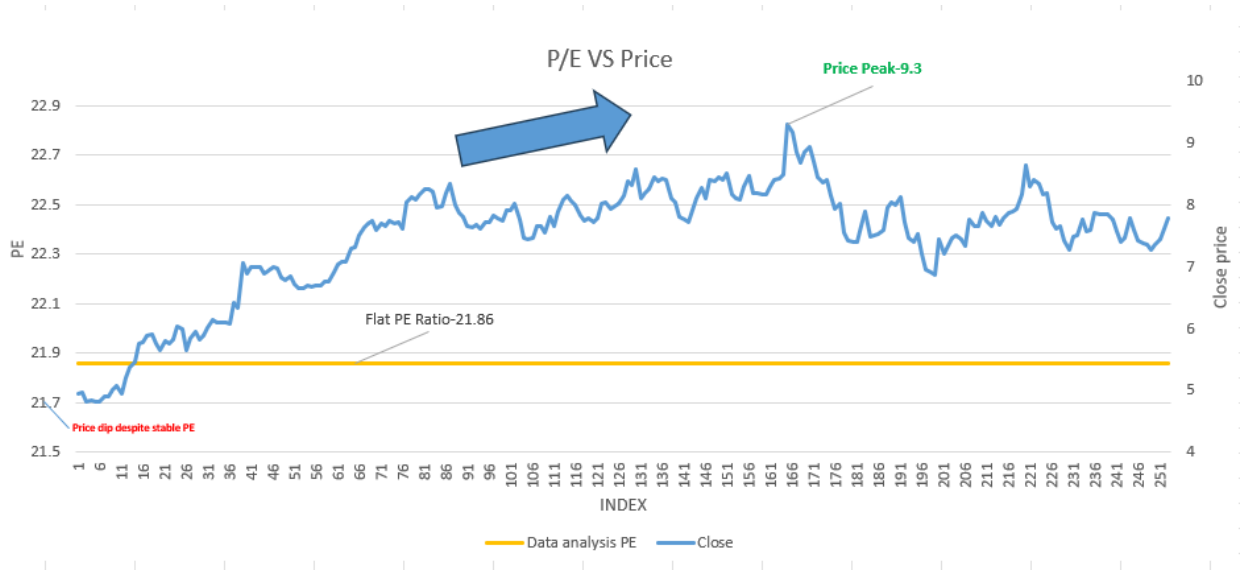
• It was clarified through the red label that the earlier reduction was preceded by a price reduction, rather than dilution.

• The peak marker marked the highest capitalization point ($713.50m). These labels and contrast of color provided context and supported narrative, so that one knew instantly what the underlying motivators of the trends were.

Challenges Encountered and Solutions

Axis scaling was one of the main issues. Market capitalization and shares on issue are scaled very differently, and if plotted together, they risked compression. This was overcome by using dual axes, so that both variables could be read clearly without data compression. Another potential source of confusion was misinterpretation. A shares-on-issue line, which was steadily flat, could paradoxically be overlooked. To counteract this, the line was presented in bold orange style, and annotations explained its significance. Lastly, the problem of data overlapping among annotations and trend lines was avoided by carefully placing labels outside crowded parts of the chart. These measures ensured that the chart conveyed the essential message: the changes in capitalization were price-driven, not dilution-driven.

4. PE Vs Price



Findings from the Data and Charts

As we can see from the chart, the UNI company's P/E ratio was constant at 21.86. The above chart shows steady earnings and valuation multiples. Despite this, the share price showed a great deal of fluctuation, peaking at 9.3 around an index of 170 after an early decline. Similarly, this divergence suggests that trading activity and external market sentiment, rather than fundamental earnings, were the driving factors of price.

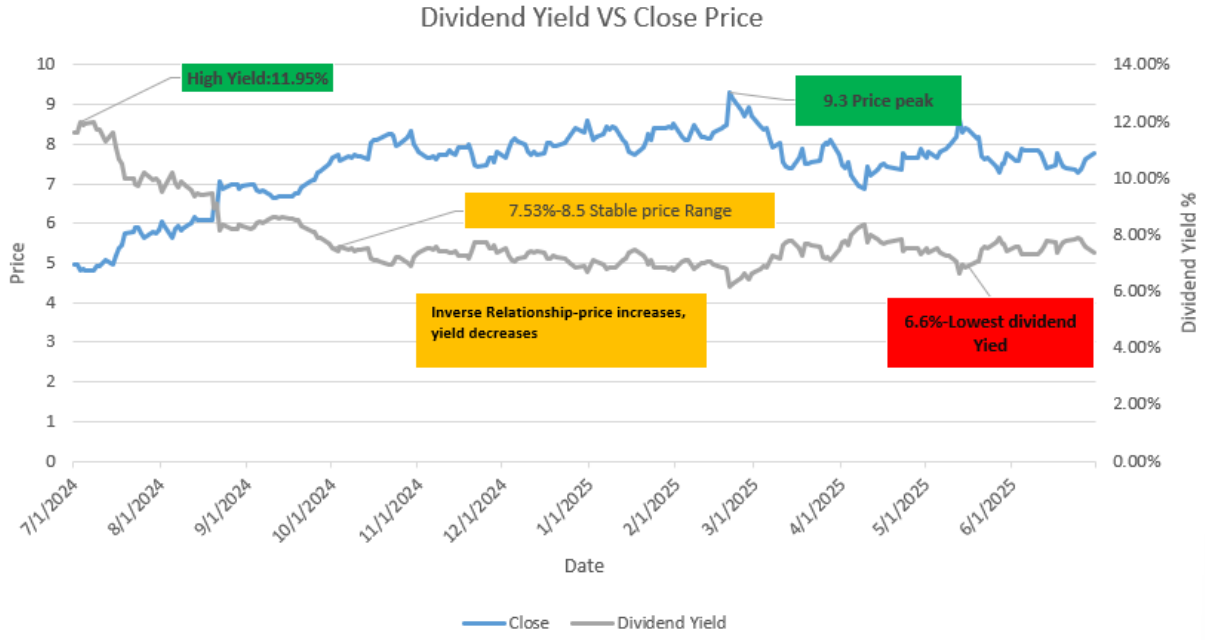
Visualization Types and Techniques

To make sure a clear contrast between the two series, a dual-axis line chart was used so that it can provide a clear comparison between the two series. This mid-price increase and early price decreases, despite stable P/E ratios, were outliers that indicated a separation between market performance and valuation. In the chart, color-coded annotations and markers, which include a red label, emphasize the price dip, whereas green label indicates the price peak. By emphasizing significant difference and enhancing interpretability, this strategy of using a line chart ensured that crucial insights were not missed.

Challenges Encountered and Solutions

The visualization technique came with several challenges. By adding contrasting color and a vivid orange line, the more animating price line was able to divert focus from the unmoving P/E ratio. Another flaw with axis scoping was that one axis would squeeze one dataset; this was rectified by employing two axes for counteracting readability. Labels and arrows were also shifted to maintain readability since annotations sometimes overlaid data lines. By implementing such shifts, the chart was successful in conveying its main argument, which was that price movement was subject to market sentiment instead of fundamentals, even though valuation multiples remained unchanged.

5. Dividend Yield Vs Close Price



Findings from the Data and Charts

The above graph shows that UNI dividend yield and share price have an inverse relationship. We can see that the dividend yield was 11.95% at the start of July 2024, while the price was lower at about $5. Likewise, the dividend yields gradually decreased as the share price increased from late 2024 to early 2025, settling at 7.56% by June 2025. This trend in the dividend graph shows that when the dividend distribution stays constant, the dividend yield falls as the price rises. Similarly, Outlier values include a sharp decline in yield in the mid 2025 and stabilization of yields around early 2025, despite fluctuations in share price. Overall, the data indicate that over time, price growth reduced the appeal of yield, making the company more appealing to investors seeking capital gains rather than those seeking income.

Visualization Types and Techniques

A dual-axis line chart was used, where the closing price is blue on the left axis and the dividend yield is grey in right axis. This chart was used to show the inverse relationship without compromising other variable .Similarly, the visual storytelling was enhanced by labelling main points such as initial yield which is 11.95% and the later low yield is 7.56%.

Challenges Encountered and Solutions

In this chart the main problem was axis scaling as dividend yield is a percentage while price is a currency. So, this problem was solved by applying dual axis which was able to keep both the variables interpretable on their respective scales. **Data crossing** was another issue: because price and yield trended in opposite directions, their lines intersected visually, risking confusion. To address this, contrasting colors and clear labels were applied to distinguish the variables. Finally, **labelling placement** required care to avoid overlapping with data lines, which was handled by positioning annotations slightly offset from the plotted series. These adjustments allowed the chart to clearly communicate the inverse yield–price dynamic and highlight how rising share prices reduce dividend-based returns.

6. Portfolio Value Over Time

In the chart the value of portfolio throughout the 2024-2025 financial year shows a dynamic evolution that shows dynamic evolution which is characterized by significant growth and elevated volatility. The Portfolio can be seen to have a period of undervaluation and low investor confidence at the beginning of July 2024, when the portfolio dropped to its lowest point of $971.72. From this base the portfolio started to have a distinct development phase, rising gradually over the following months. This upward trend can be seen to be continued through late 2024 which is supported by stronger price movements gradually increasing overall value. Similarly, the portfolio nearly doubled in value from the low point in a short span of time as the portfolio reached a maximum of $1878.70 in March 2025, marking the end of bull market.

While there was a notable growth rising trajectory, the chart also shows a series of significant volatility. The Portfolio can be seen to have rapid swings, with steep declines followed by partial recoveries. This makes it evident that even though the stock had a huge potential to yield significant profits, it was also vulnerable to short term uncertainty .Despite the challenges, the dotted trend lines on the chart supports the portfolio with upward trajectory which demonstrates that long term trajectory held steady even during disruptive times.

Likewise, the visualization effectively communicates these findings using different annotations and color coding. A green arrow in the graph shows the growth phase, directing the viewer's attention to a period of steady upward movement, while a red arrow in the graph shows high

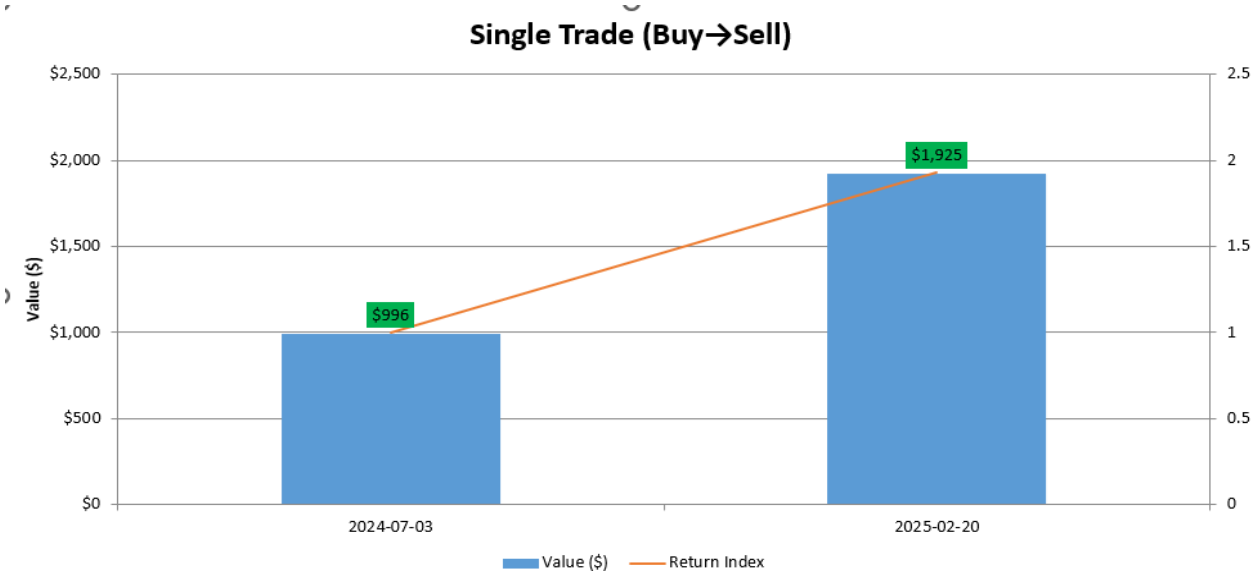
market volatility section to emphasis the risk involved. To make the crucial outliers instantly visible, the lowest and highest points are clearly marked with the peak in green and low in red text. In addition to improving readability, these labelling and visual strategies enable the chart to function as a storytelling tool, assisting the audience in following the story of rise, collapse, and long-term development.

Overall, the portfolio value overtime analysis demonstrates that the investment performed strongly throughout the financial year, nearly doubling in value from its lowest to highest point. It shows that even though there was a time when investors profited from growth, there was also a time when there was a considerable risk. So,the chart by combining all raw data with targeted annotations can capture both the opportunities and challenges inherent in the portfolio's journey providing accessible and compelling account of performance across 2024–2025

7. Single Trade analysis

The analysis of a single trade gives perspective on the success of a carefully timed buy-and-sell approach during the financial year 2024–2025.The first chart, which illustrates daily closing prices with a trendline, outlines the stock's trajectory between the chosen to buy and sell points. The purchase was made on 3 July 2024 at a value of $4.81, allowing for 207 shares to be bought.

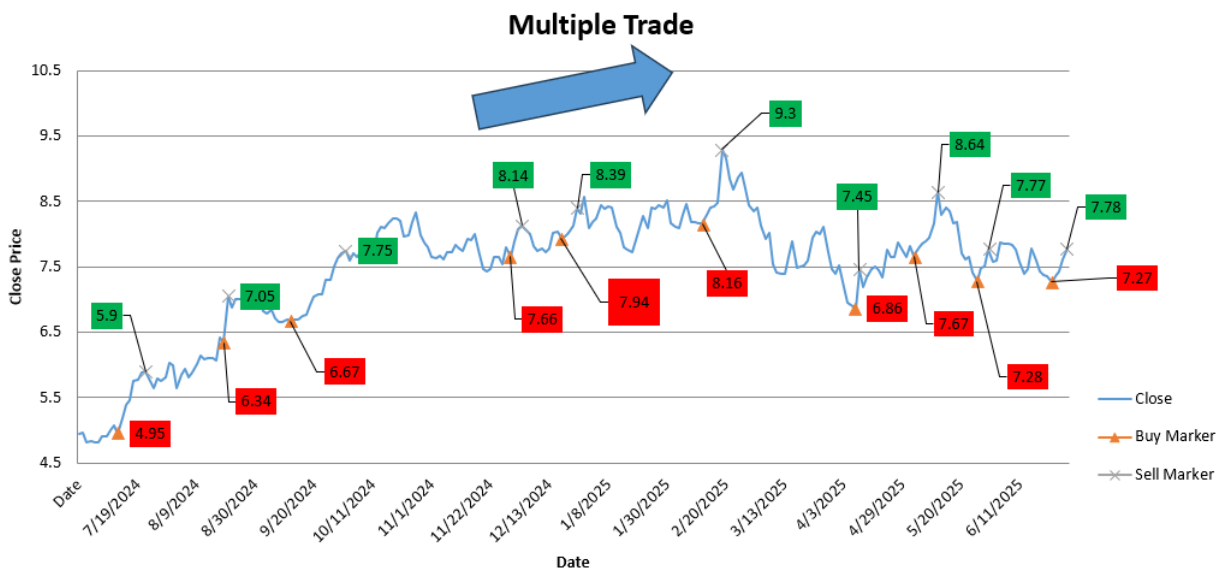
This point of view was a strategic choice, as the price neared the year's lower limit, thus offering considerable upside room. The portfolio was then held through a bull trend, ending with sales on 20 February 2025 at a value of $9.30. This exit realized the nearest peak of the stock's movement, reaching a final figure of $1,925.10 from the initial investment of $995.67. The profit derived from this single trade amounted to $929.43, representing a return of approximately 93.35 percent. The trendline shows upward momentum throughout the holding period, with the trendline demonstrating consistent growth throughout. Similarly, the buy point and the high point of the upward trend are both shown in the chart with the help of annotations which provides a clear visual to follow along the trade process. Likewise, the labelling in the chart shows clear reasoning behind the trade-purchase during undervaluation and sale at close to the top of an upward trend—and the visual clarity of the upward trend supports the success of the trade and its timing

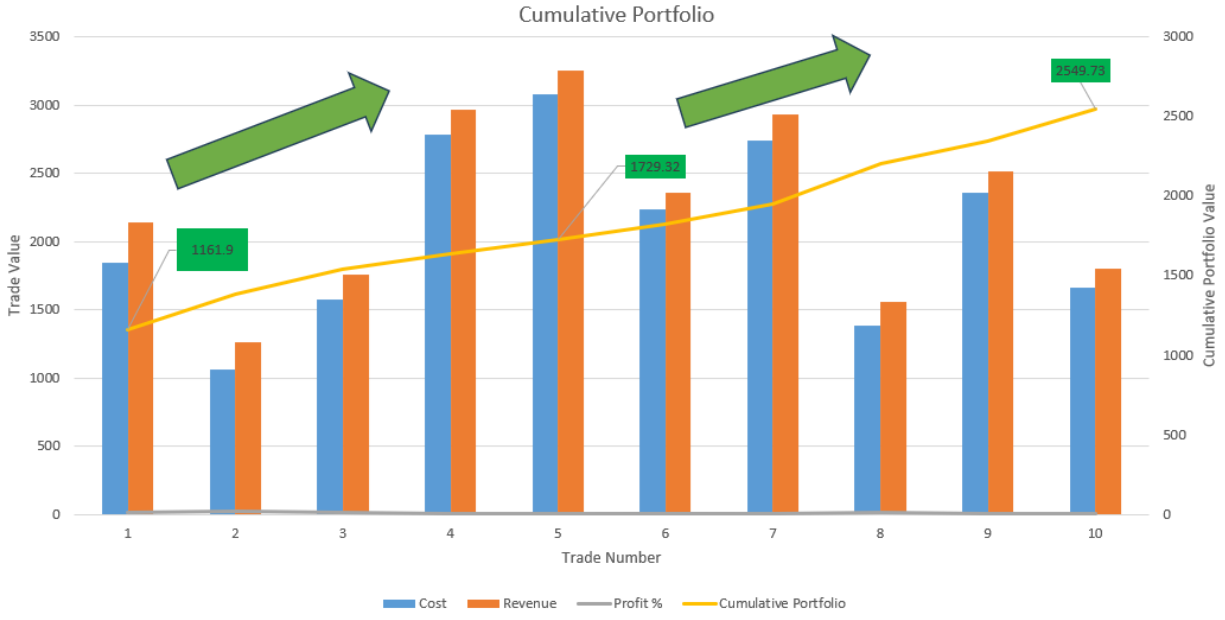


The second chart displays the outcomes of trade, displaying how much value increased from beginning to end. The bar chart displays the growth from $995.67 to $1,925.10, and the return index line displays the percentage growth, beginning at 1.0 (100 percent) and ending at around 1.93 (193 percent of the original value). This method of displaying value and percentage growth makes it easier for people to view the outcome in two ways: the end value and the percentage growth. Data labels on the bars assist by displaying the actual end-of-day dollar amounts at the ends.

The two adjoining graphs demonstrate the worth of the single good rule. They reveal that having a single good decision can almost double the initial money in the long run. Meanwhile, the graphs reflect that the process depends greatly on timeliness because slight alterations in when to sell or buy may immensely reduce the aggregated gain. This implies that personal transactions earn a lot of money but are unsafe and depend on having the capability of forecasting or responding well to changes in the market

8. Multiple Trade analysis





Multiple trade study displays an evident comprehension of active trading strategy's potential to optimize portfolio performance over the span of time. Cumulative portfolio chart illustrates the portfolio's growth from its initial $1,000 value to $2,549.73 during financial year end in a stream of ten successive trades. As a return of almost 155%, the outcome is far more spectacular than an outcome of a lone one trade or buy and hold approach. The consistent upward ascent of the yellow aggregate portfolio's line through the trades illustrates the way incomes of reinvestment into subsequent trades worked to compound the growth. Chart annotations emphasize noteworthy changes taking place along the way.
The portfolio was over $1,161.90 following the first transaction, $1,729.32 following the fifth transaction, and $2,549.73 following the tenth transaction. Cost and revenue bars and each transaction also show the sum of money invested and put into perspective when returns have been generated by intermittent buy and sell cycles.

The second graph illustrates all the buy and sell actions that occurred throughout the year. There are indicators on each trade that indicate when to sell and buy. As an example, shares were purchased for $4.95 and sold subsequently for $5.90, profiting on the initial trade. The graph illustrates the same information, with green labels illustrating where sales were profitable and red labels illustrating where goods were sold for a loss. The graph illustrates not only how the trades were organized, but illustrates market fluctuations, as prices rose and fell between lows of roughly $4.95 and highs of $9.30 throughout the year. By reacting to these fluctuations, the strategy profited from short-term opportunity rather than relying solely on long-term growth.

These charts collectively convey how various trades contributed to the growth of the portfolio. The cumulative chart illustrates the total growth of the funds from compounding, and the transaction-level chart illustrates the choices made to bring about the growth. The arrows, labels, and colored markers make it easy and obvious how a particular trade impacted the overall outcome.

The study reveals that active trading requires more effort and labor, but it will result in far better profits than less frequent trading. It is also evidenced that having a sense of timing, observing the screen intently, and an understanding that the prices will rise and fall are extremely crucial. Both of the trades incurred risks, indicated by the red symbols for declining sale prices, and poor

timing may have damaged the development of the portfolio. However, the plan performed well here and produced the best outcome of all the plans considered.

Conclusion and Recommendations

An analysis of UNI company's stock performance between 2024 and 2025 can be seen to have both opportunities and risks for potential investors' buy and hold strategy produced high long-term results but exposed the investors to severe market volatility. A single trade which can be seen to be timed at low entry and peaked near exit, which had a return of nearly double the initial investment, demonstrating that when short term strategies are used with precision gives profitable returns. Likewise, multiple trade can be seen to give the best return of around 150% by compounding profits over different transactions. Nevertheless, the method needed constant monitoring precise timing and a tolerance of short-term fluctuations

With the help of these comparisons, investors who are seeking stability may favor a long-term holding strategy but those who have higher market knowledge and higher risk tolerance can gain from trade-based strategies. The finding indicates that a diversified approach which combines long term hold with selected active trading, may deliver the most balanced result for future investors.

An important aspect of this report was the use of different visualization tools for decision making. Different charts like candlestick plots, dual axis charts and annotated graphs were used to transform complex financial trends into understandable stories for buyers and investors. Different types of techniques like color coding, arrows and highlighted outliers made volume-driven price peaks, inverse yield–price relationships, and portfolio growth instantly interpretable. Unlike AI which often provides charts without tailoring them to specific investor needs and may simplify chart way too much, leaving out annotations and storytelling elements. Similarly, it may provide overly complex visualization which may make the interpretation harder, and they can't provide investor focused story telling.

Overall, this report demonstrates how thoughtful data preparation and visualization can effectively communicate crucial financial information, enabling investors to make correct decisions in complication and volatile markets.